

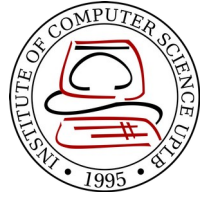
CMSC 137

Data Communications and Networking

Lecture Module
03 – TCP/IP Protocol Suite

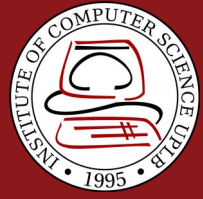


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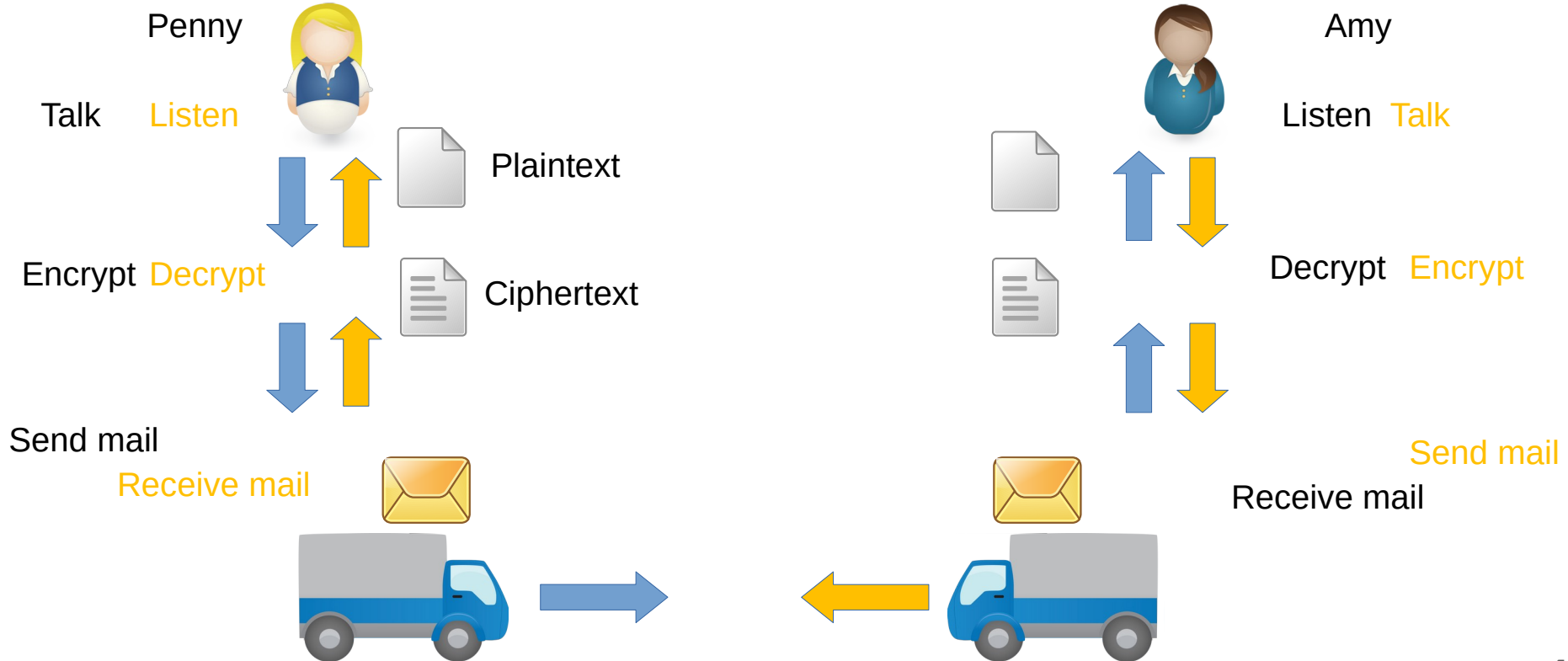
Module 3: TCP/IP Protocol Suite

Rozano S. Maniaol
Lecturer

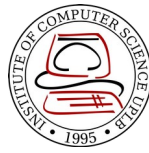


Protocol Layering

Sending a message

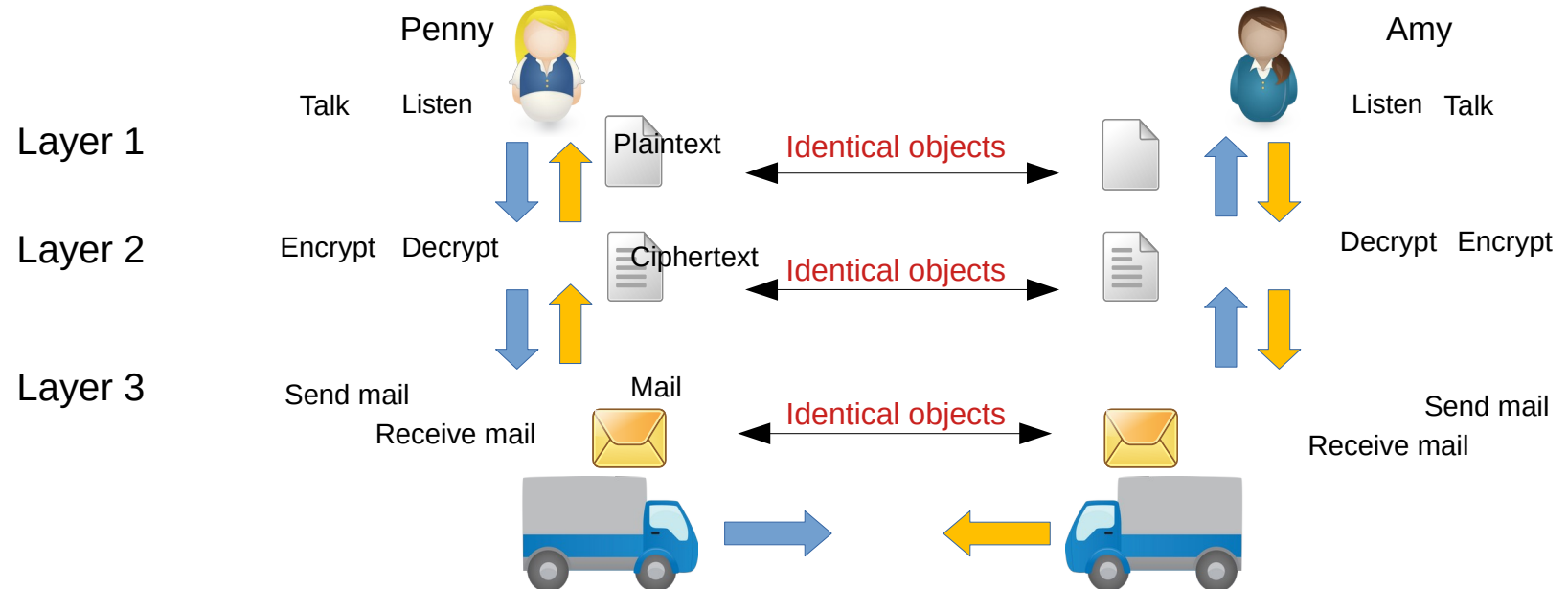


Protocol Layering

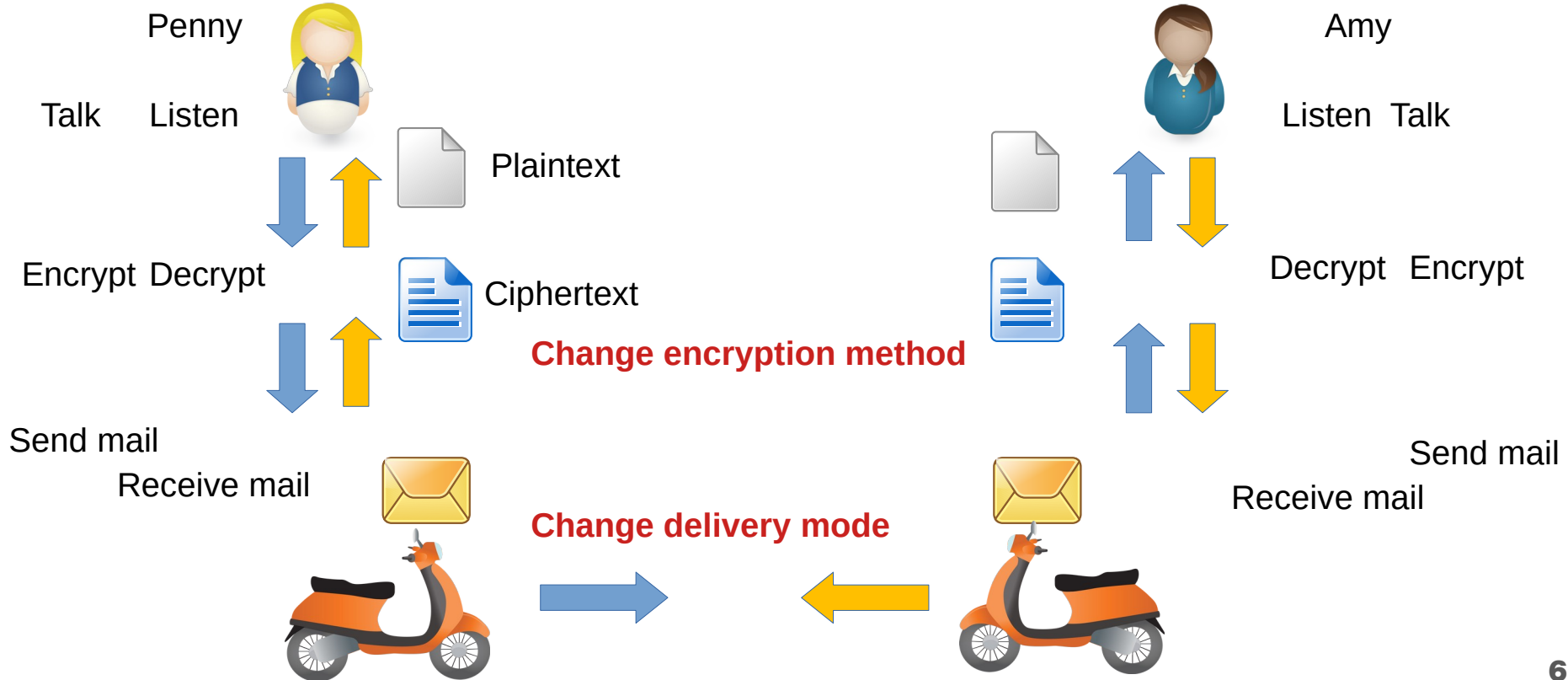


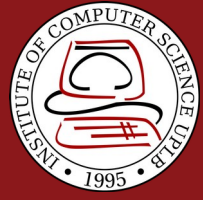
Principles:

- Each layer needs to have two opposite tasks
- Corresponding layers should be identical



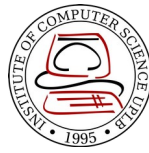
Sending a message



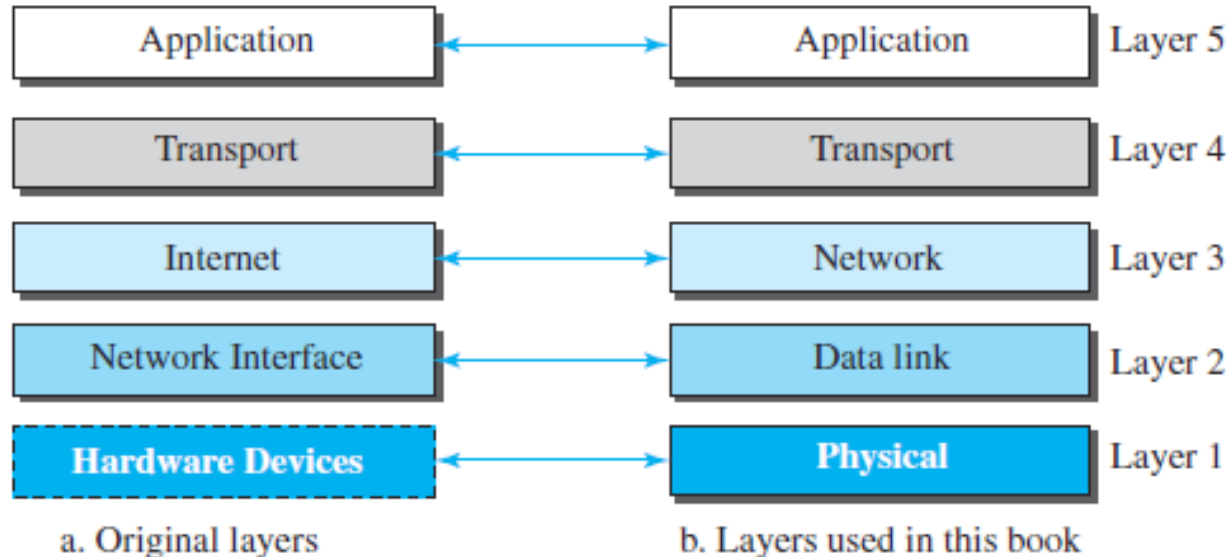


TCP/IP Protocol Suite

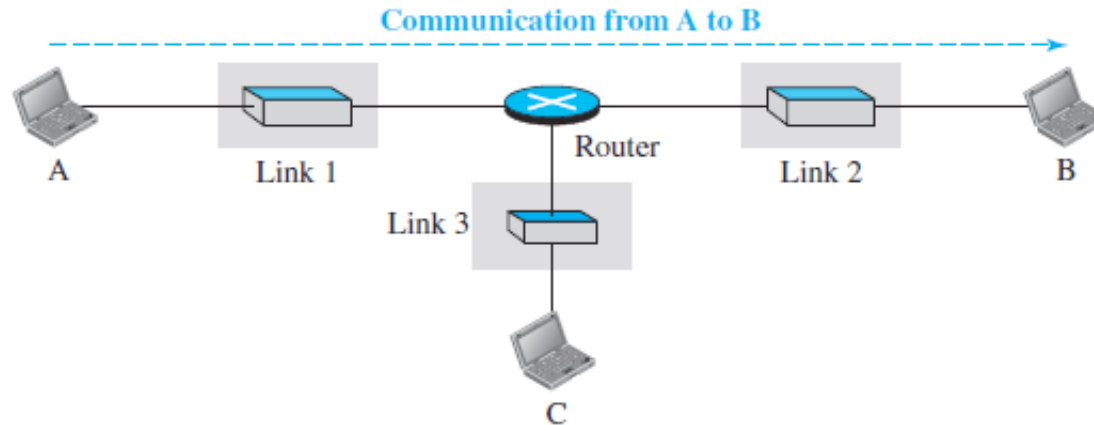
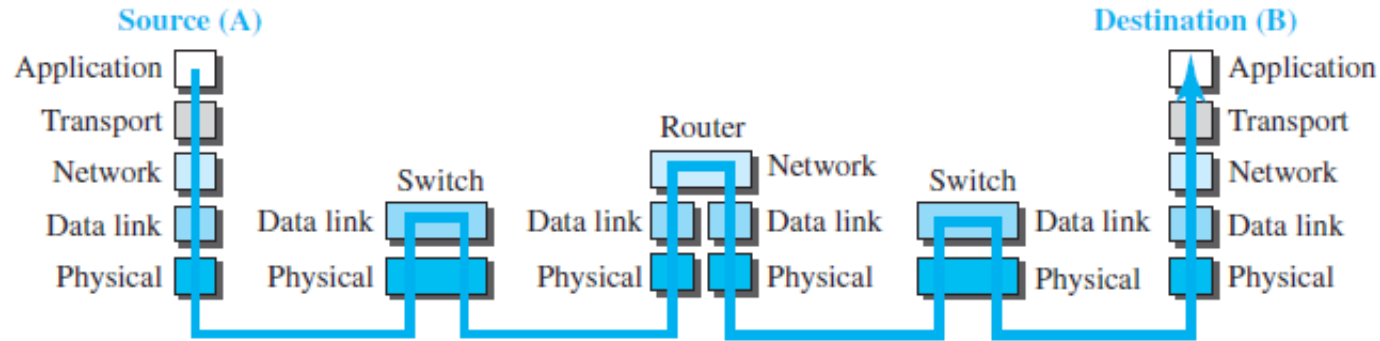
TCP/IP Protocol Suite



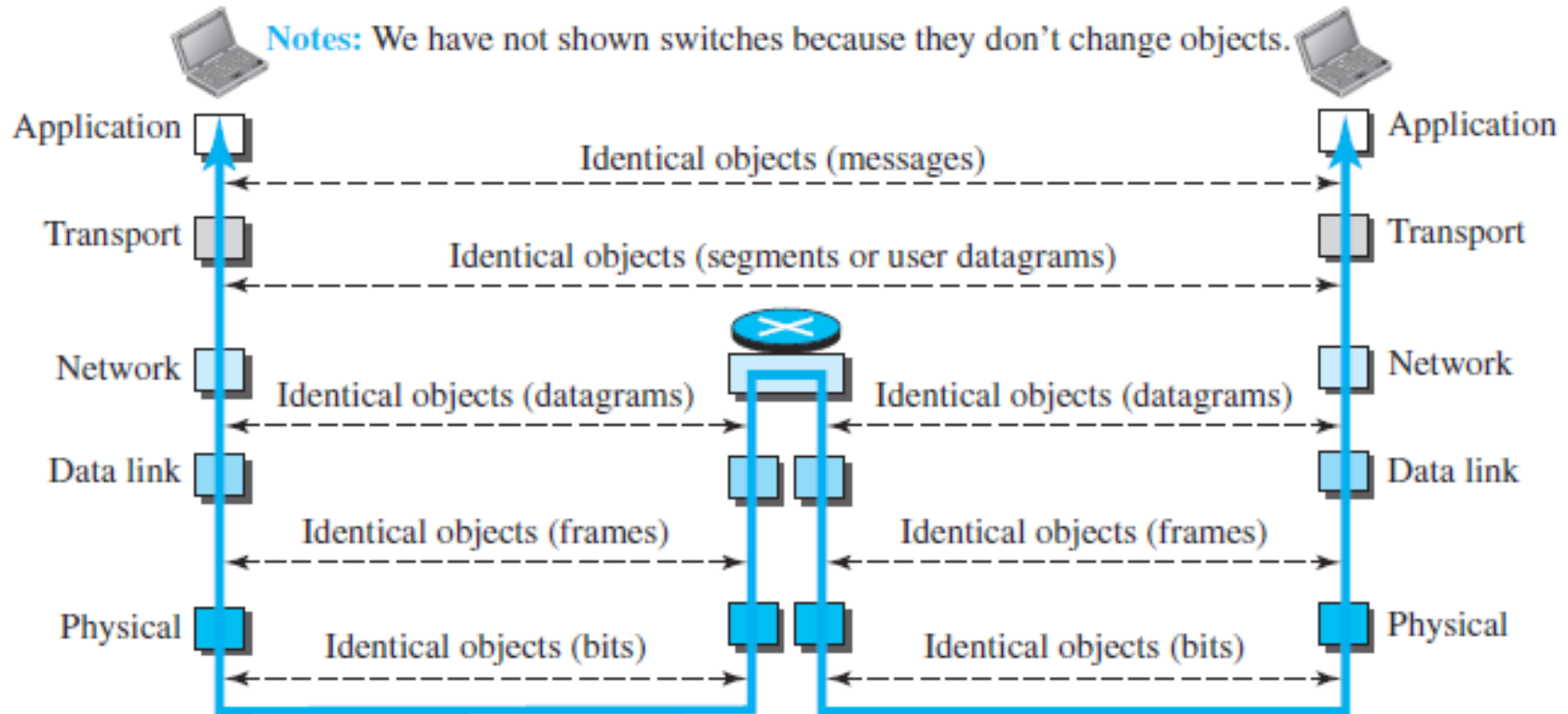
- Hierarchical protocol made up of interactive modules – upper-level protocol is supported by one or more lower-level protocols
- Also called the Internet Protocol Suite



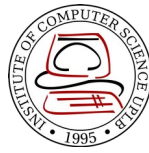
TCP/IP Protocol Suite



Identical Objects



Physical Layer



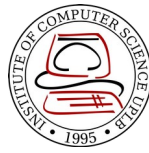
- Identifies the characteristics of the hardware to be used for the network
- Responsible for carrying individual bits in a frame across the link
- Communication is still logical (bits); a hidden layer underneath (transmission media) carries electrical/optical signals

Data Link Layer



- Identifies the network protocol type of the packet
- TCP/IP does not define any specific protocol for data link layer. It supports all the standard and proprietary protocols

Network Layer



- Responsible for host-to-host communication and routing the packets through possible routes
- **Internet Protocol (IP)** [**RFC 791**]
 - Defines the format of the packet (datagram)
 - Defines the format and structure of the addresses used in this layer
 - Connectionless protocol that provides no flow control, no error control and no congestion control services

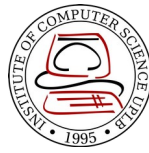
Network Layer



Auxiliary protocols

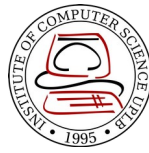
- [Internet Control Message Protocol](#) (ICMP) – helps IP to report some problems when routing a packet
- [Internet Group Management Protocol](#) (IGMP) – helps IP in multitasking
- [Dynamic Host Configuration Protocol](#) (DHCP) – helps IP to get network-layer address for a host
- [Address Resolution Protocol](#) (ARP) – helps IP find the link-layer address of a host or a router when its network-layer address is given

Transport Layer



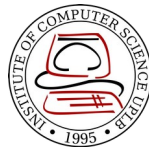
- Responsible for giving services to the application layer
- **Transmission Control Protocol (TCP)** [**RFC 793**]
 - Connection-oriented protocol that first established a logical connection between transport layers at 2 host before transferring data
 - *flow control* – match sending data rate of the source host with receiving data rate of the destination to prevent from overwhelming the destination
 - *error control* – guarantee that segments arrive at the destination without error and resending the corrupted ones
 - *congestion control* – to reduce the loss of segments due to the congestion in the network

Transport Layer



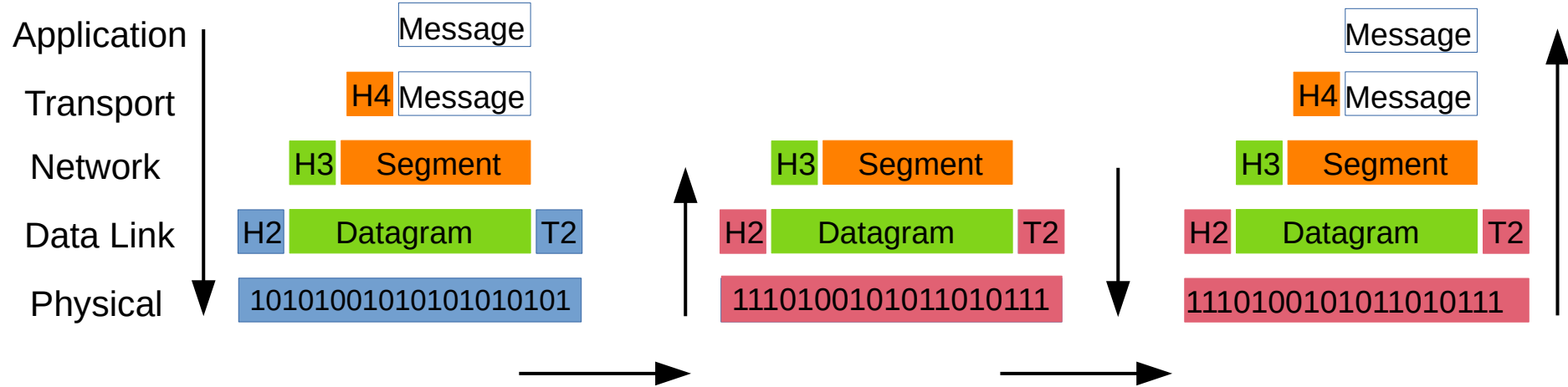
- **User Datagram Protocol (UDP)** [RFC 768]
 - Connectionless protocol that transmits user datagrams without first creating a logical connection
 - Adds only port addresses, checksum error control, and length information
- **Stream Control Transmission Protocol (SCTP)** [RFC 4960]
 - Best of TCP and UDP
 - Designed to respond to new applications that are emerging in the multimedia

Application Layer

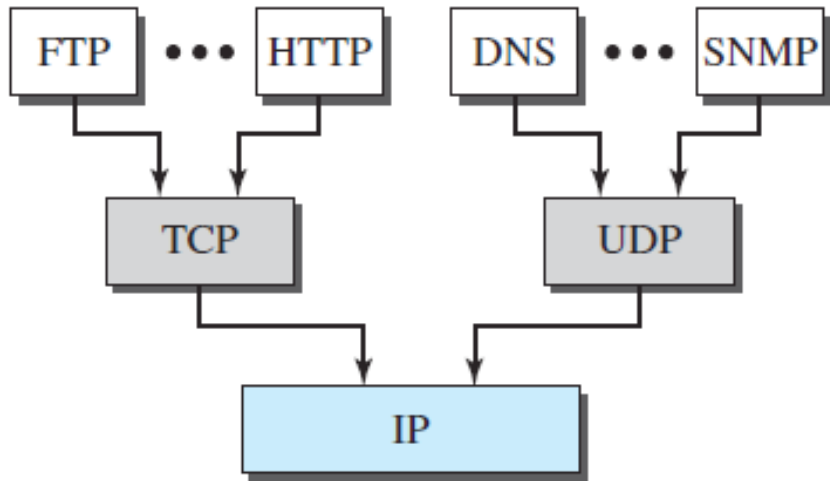


- Process to process communication
- [Hypertext Transfer Protocol](#) (HTTP) – vehicle for accessing the World Wide Web (WWW)
- [Simple Mail Transfer Protocol](#) (SMTP) – main protocol used in electronic mail service
- [File Transfer Protocol](#) (FTP) – used for transferring files from one host to another
- [Terminal Network](#) (Telnet) and [Secure Shell](#) (SSH) – used for accessing remote sites
- [Simple Network Management Protocol](#) (SNMP) – used by an administrator to manage the Internet at global and local levels

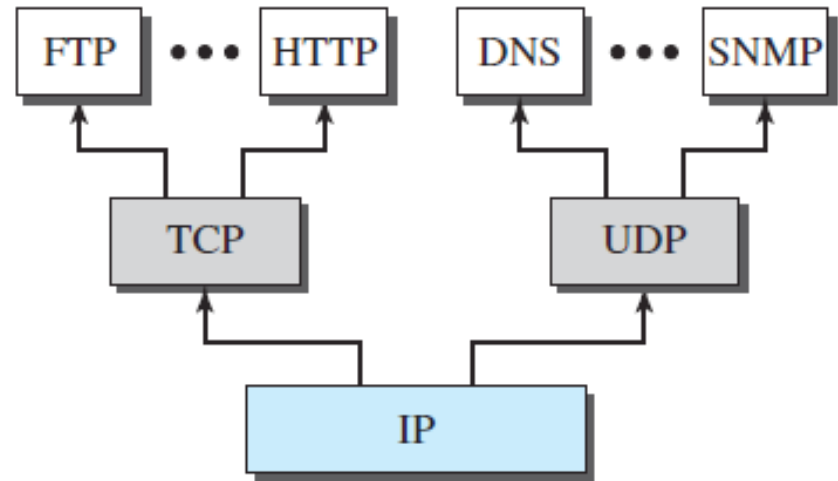
Encapsulation and Decapsulation



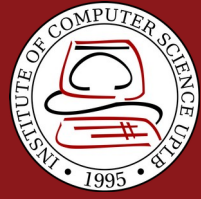
Multiplexing and Demultiplexing



a. Multiplexing at source



b. Demultiplexing at destination



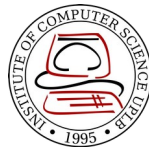
Addressing

TCP/IP Addressing



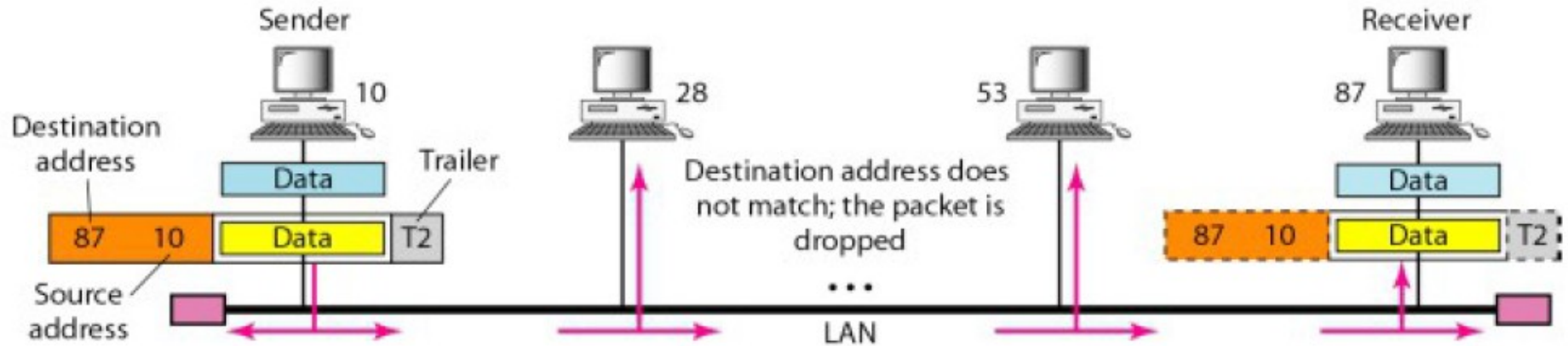
Packet names	Layers	Addresses
Message	Application layer	Names
Segment / User datagram	Transport layer	Port numbers
Datagram	Network layer	Logical addresses
Frame	Data-link layer	Link-layer addresses
Bits	Physical layer	

Link-layer Address

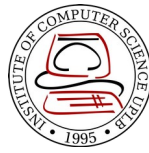


- Also called **physical address** or **MAC address**
- Lowest-level address defined by a node's LAN or WAN
- Size and format varies depending on network
 - Example: Ethernet uses 6-byte physical addresses written as 12 hexadecimal digits separated by a colon
 - 68:a3:c4:ce:8c:e2
- **Organizationally Unique Identifier (OUI)**
 - Dell: 00-14-22
 - Nortel: 00-04-DC
 - Cisco: 00-40-96
 - Belkin: 00-30-BD

Link-layer Address

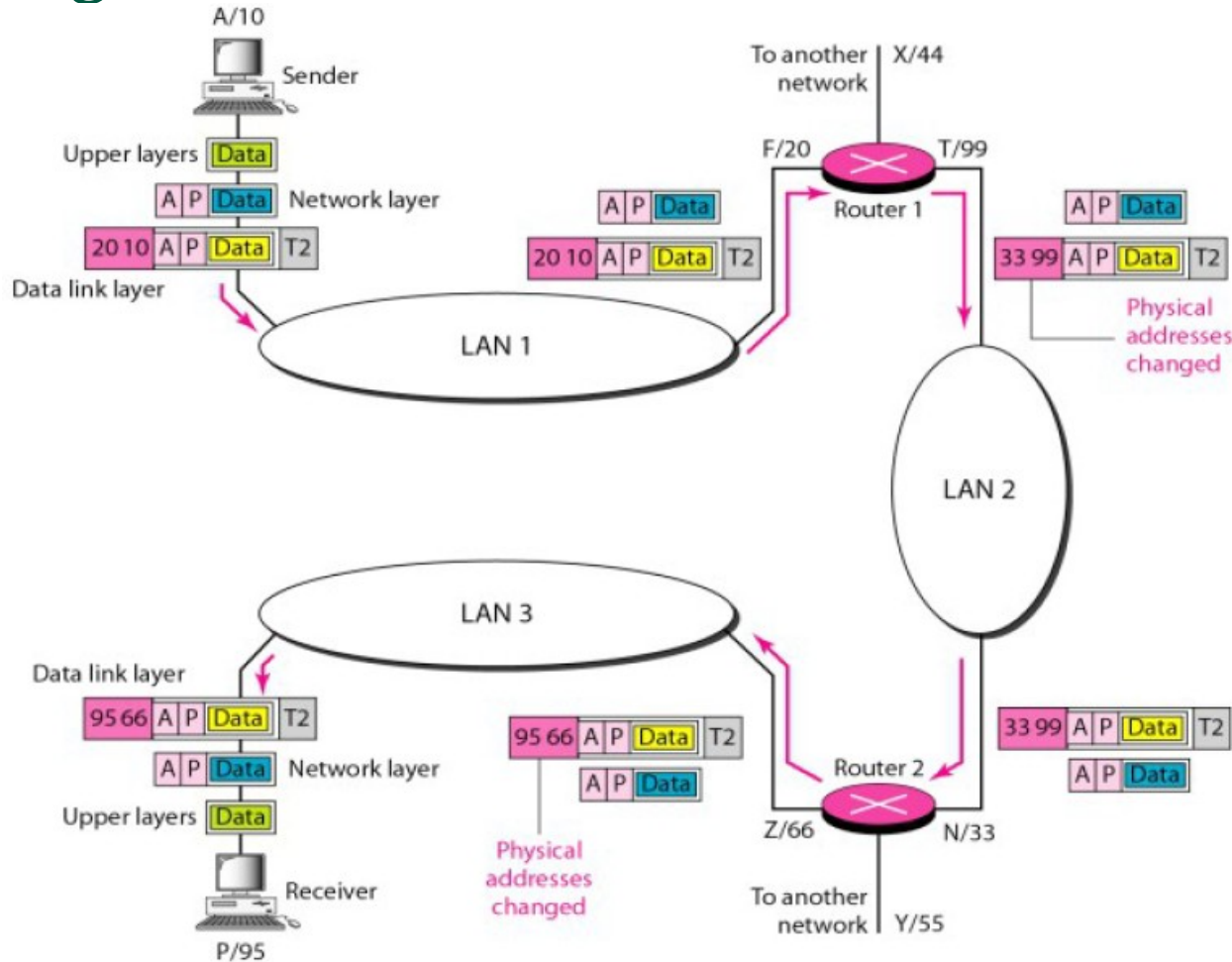


Logical Address



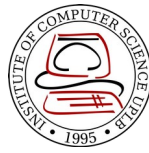
- Necessary for **universal** communications – internetworks
- Internet (IP) address is 32 bits
- No two publicly addressed and visible host on the Internet can have the same logical address

Logical Address



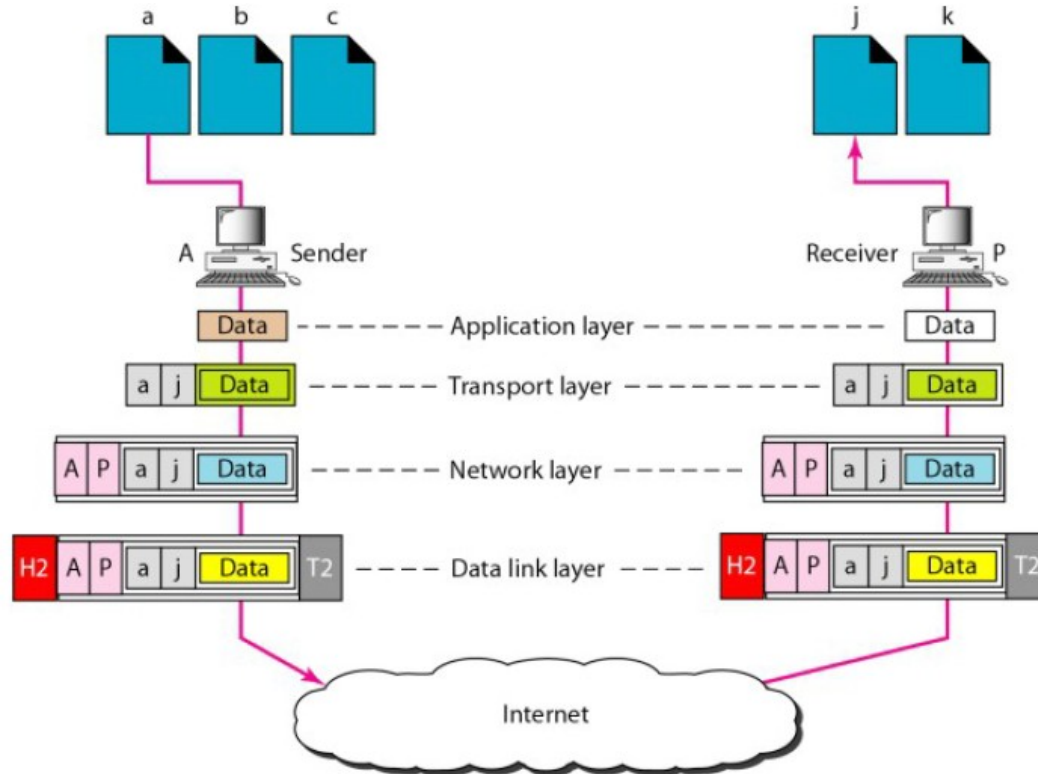
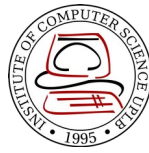
The physical addresses (numbers) changes between LANs but the logical addresses (letters) does not

Port Number



- Ultimate objective of Internet communication is a process communicating with another (peer) process running on a remote computer
- TCP/IP Port address – assigned to a process, 16 bits
- Examples: 80, 23, 22, 21, 8080

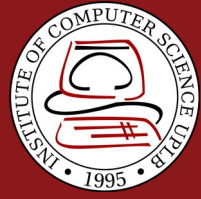
Port Number



Names/Specific Names



- User-friendly addresses
 - Example: email addresses, URL
- Changed to corresponding port and logical addresses



How it works

Sample Packet

No.	Time	Source	Destination	Protocol	Length	Info
539	13.616736000	192.168.1.2	95.172.71.48	HTTP	794	GET / HTTP/1.1
561	14.028335000	192.168.1.2	95.172.71.48	HTTP	728	GET /wp-content/plugins/easy-fancybox/easy-fancybox.css.php?ver=1.3.4 HTTP/1.1
569	14.083458000	192.168.1.2	95.172.71.48	HTTP	727	GET /wp-content/uploads/jw-player-plugin-for-wordpress/player/jwplayer.js?ver=4.2.4 HTTP/1.1
623	14.340281000	95.172.71.48	192.168.1.2	HTTP	524	HTTP/1.1 200 OK (text/html)
634	14.346269000	192.168.1.2	74.125.130.95	HTTP	621	GET /css?family=open+Sans:700,800,400 HTTP/1.1
644	14.405804000	95.172.71.48	192.168.1.2	HTTP	589	HTTP/1.1 301 Moved Permanently (text/html)
+ Frame 539: 794 bytes on wire (6352 bits), 794 bytes captured (6352 bits) on interface 0						
+ Ethernet II, Src: HonHaiPr_6e:93:f7 (e0:06:e6:93:f7), Dst: Zte_49:31:3d (34:4b:50:49:31:3d)						
+ Internet Protocol Version 4, Src: 192.168.1.2 (192.168.1.2), Dst: 95.172.71.48 (95.172.71.48)						
+ Transmission Control Protocol, Src Port: 53062 (53062), Dst Port: 80 (80), Seq: 1, Ack: 1, Len: 740						
+ Hypertext Transfer Protocol						
+ GET / HTTP/1.1\r\n						
Host: www.inquirer.net\r\n						
Connection: keep-alive\r\n						
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8\r\n						
Upgrade-Insecure-Requests: 1\r\n						
User-Agent: Mozilla/5.0 (windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/44.0.2403.157 Safari/537.36\r\n						
DNT: 1\r\n						
Accept-Encoding: gzip, deflate, sdch\r\n						
Accept-Language: en-US,en;q=0.8\r\n						
<						
0000	34 4b 50 49 31 3d e0 06	e6 6e 93 f7 08 00 45 00	4KPII=...n....E.			
0010	03 0c 60 73 40 00 80 06	2e f2 c0 a8 01 02 5f ac	..`s@...			
0020	47 30 cf 46 00 50 b1 32	b9 81 32 27 c9 61 50 18	G0.F.P.2 ..2'.ap.			
0030	01 01 c8 ed 00 00 47 45	54 20 2f 20 48 54 54 50	GE T / HTTP			
0040	2f 31 2e 31 0d 0a 48 6f	73 74 3a 20 77 77 77 2e	/1.1..Host: www.			
0050	69 6e 71 75 69 72 65 72	2e 6e 65 74 0d 0a 43 6f	inquirer .net..co			
0060	6e 6e 65 63 74 69 6f 6e	3a 20 6b 65 65 70 2d 61	nnection : keep-a			
0070	6c 69 76 65 0d 0a 41 63	63 65 70 74 3a 20 74 65	live..Accept: te			
0080	78 74 2f 68 74 6d 6c 2c	61 70 70 6c 69 63 61 74	xt/html, applicat			
0090	69 6f 6e 2f 78 68 74 6d	6c 2b 78 6d 6c 2c 61 70	ion/xhtml l+xml,ap			
00a0	70 6c 69 63 61 74 69 6f	6e 2f 78 6d 6c 3b 71 3d	plicatio n/xml;q=			
00b0	30 2e 39 2c 69 6d 61 67	65 2f 77 65 62 70 2c 2a	0.9,image/webp,*			
00c0	2f 2a 3b 71 3d 30 2e 38	0d 0a 55 70 67 72 61 64	/*;q=0.8 ..Upgrad			
00d0	65 2d 49 6e 73 65 63 75	72 65 2d 52 65 71 75 65	e-Insecu re-Reque			
00e0	73 74 73 3a 20 31 0d 0a	55 73 65 72 2d 41 67 65	sts: 1.. User-Age			
00f0	6e 74 3a 20 4d 6f 7a 69	6c 6c 61 2f 35 2e 30 20	nt: Mozi lla/5.0			
0100	28 57 69 6e 64 6f 77 73	20 4e 54 20 31 30 2e 30	(windows NT 10.0			
0110	7b 30 57 4f 57 75 74 30	70 41 70 70 65 57 65	* WOW64) AppleWeb			

HTTP Packet (Application Layer)

No.	Time	Source	Destination	Protocol	Length	Info
539	13.616736000	192.168.1.2	95.172.71.48	HTTP	794	GET / HTTP/1.1
561	14.028335000	192.168.1.2	95.172.71.48	HTTP	728	GET /wp-content/plugins/easy-fancybox/easy-fancybox.css.php?ver=1.3.4 HTTP/1.1
569	14.083458000	192.168.1.2	95.172.71.48	HTTP	727	GET /wp-content/uploads/jw-player-plugin-for-wordpress/player/jwplayer.js?ver=4.2.4 HTTP/1.1
623	14.340281000	95.172.71.48	192.168.1.2	HTTP	524	HTTP/1.1 200 OK (text/html)
634	14.346269000	192.168.1.2	74.125.130.95	HTTP	621	GET /css?family=Open+Sans:700,800,400 HTTP/1.1
644	14.405804000	95.172.71.48	192.168.1.2	HTTP	589	HTTP/1.1 301 Moved Permanently (text/html)

Frame 539: 794 bytes on wire (6352 bits), 794 bytes captured (6352 bits) on interface 0

Ethernet II, Src: HonHaiPr_6e:93:f7 (e0:06:e6:6e:93:f7), Dst: Zte_49:31:3d (34:4b:50:49:31:3d)

Internet Protocol Version 4, Src: 192.168.1.2 (192.168.1.2), Dst: 95.172.71.48 (95.172.71.48)

Transmission Control Protocol, Src Port: 53062 (53062), Dst Port: 80 (80), Seq: 1, Ack: 1, Len: 740

Hypertext Transfer Protocol

GET / HTTP/1.1\r\nHost: www.inquirer.net\r\nConnection: keep-alive\r\nAccept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8\r\nUpgrade-Insecure-Requests: 1\r\nUser-Agent: Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/44.0.2403.157 Safari/537.36\r\nDNT: 1\r\nAccept-Encoding: gzip, deflate, sdch\r\nAccept-Language: en-US,en;q=0.8\r\n

0000 34 4b 50 49 31 3d e0 06 e6 6e 93 f7 08 00 45 00 4KPI1=...n...E.

0010 03 0c 60 73 40 00 80 06 2e f2 c0 a8 01 02 5f ac ..s@...

0020 47 30 cf 46 00 50 b1 32 b9 81 32 27 c9 61 50 18 G0.F.P.2 ..2'.aP.

0030 01 01 c8 ed 00 00 47 45 54 20 2f 20 48 54 54 50GE T / HTTP

0040 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 77 77 77 2e /1.1..Ho st: www.

0050 69 6e 71 75 69 72 65 72 2e 6e 65 74 0d 0a 43 6f inquirer .net..Co

0060 6e 6e 65 63 74 69 6f 6e 3a 20 6b 65 65 70 2d 61 nnection : keep-a

0070 6c 69 76 65 0d 0a 41 63 63 65 70 74 3a 20 74 65 live..Ac cept: te

0080 78 74 2f 68 74 6d 6c 2c 61 70 70 6c 69 63 61 74 xt/html, applicat

0090 69 6f 6e 2f 78 68 74 6d 6c 2b 78 6d 6c 2c 61 70 ion/xhtml l+xml,ap

00a0 70 6c 69 63 61 74 69 6f 6e 2f 78 6d 6c 3b 71 3d plicatio n/xml;q=

00b0 30 2e 39 2c 69 6d 61 67 65 2f 77 65 62 70 2c 2a 0.9,imag e/webp,*

00c0 2f 2a 3b 71 3d 30 2e 38 0d 0a 55 70 67 72 61 64 /*;q=0.8 ..Upgrad

00d0 65 2d 49 6e 73 65 63 75 72 65 2d 52 65 71 75 65 e-Insecu re-Reque

00e0 73 74 73 3a 20 31 0d 0a 55 73 65 72 2d 41 67 65 sts: 1.. user-Age

00f0 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 nt: Mozi lla/5.0

0100 28 57 69 6e 64 6f 77 73 20 4e 54 20 31 30 2e 30 (windows NT 10.0

0110 2b 20 52 4f 52 36 74 20 20 41 70 20 6e 65 52 65)

31

TCP Header (Transport Layer)

No.	Time	Source	Destination	Protocol	Length	Info
539	13.616736000	192.168.1.2	95.172.71.48	HTTP	794	GET / HTTP/1.1
561	14.028335000	192.168.1.2	95.172.71.48	HTTP	728	GET /wp-content/plugins/easy-fancybox/easy-fancybox.css.php?ver=1.3.4 HTTP/1.1
569	14.083458000	192.168.1.2	95.172.71.48	HTTP	727	GET /wp-content/uploads/jw-player-plugin-for-wordpress/player/jwplayer.js?ver=4.2.4 HTTP/1.1
623	14.340281000	95.172.71.48	192.168.1.2	HTTP	524	HTTP/1.1 200 OK (text/html)
634	14.346269000	192.168.1.2	74.125.130.95	HTTP	621	GET /css?family=open+Sans:700,800,400 HTTP/1.1
644	14.405804000	95.172.71.48	192.168.1.2	HTTP	589	HTTP/1.1 301 Moved Permanently (text/html)

Frame 539: 794 bytes on wire (6352 bits), 794 bytes captured (6352 bits) on interface 0
Ethernet II, Src: HonHaiPr_6e:93:f7 (e0:06:e6:6e:93:f7), Dst: Zte_49:31:3d (34:4b:50:49:31:3d)
Internet Protocol Version 4, Src: 192.168.1.2 (192.168.1.2), Dst: 95.172.71.48 (95.172.71.48)
Transmission Control Protocol, Src Port: 53062 (53062), Dst Port: 80 (80), Seq: 1, Ack: 1, Len: 740
Hypertext Transfer Protocol
GET / HTTP/1.1\r\n
Host: www.inquirer.net\r\n
Connection: keep-alive\r\n
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8\r\n
Upgrade-Insecure-Requests: 1\r\n
User-Agent: Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/44.0.2403.157 Safari/537.36\r\n
DNT: 1\r\n
Accept-Encoding: gzip, deflate, sdch\r\n
Accept-Language: en-US,en;q=0.8\r\n

0000	34 4b 50 49 31 3d e0 06	e6 6e 93 f7 08 00 45 00	4KPI1=.. .n...E.
0010	03 0c 60 73 40 00 80 06	2e f2 c0 a8 01 02 5f ac	.. s@...
0020	47 30 cf 46 00 50 b1 32	b9 81 32 27 c9 61 50 18	G0.F.P.2 ..2'.aP
0030	01 01 c8 ed 00 00 47 45	54 20 2f 20 48 54 54 50	GET / HTTP
0040	2f 31 2e 31 0d 0a 48 6f	73 74 3a 20 77 77 77 2e	/1.1..Host: www.
0050	69 6e 71 75 69 72 65 72	2e 6e 65 74 0d 0a 43 6f	inquirer .net..Co
0060	6e 6e 65 63 74 69 6f 6e	3a 20 6b 65 65 70 2d 61	nnnection : keep-a
0070	6c 69 76 65 0d 0a 41 63	63 65 70 74 3a 20 74 65	live..Ac cept: te
0080	78 74 2f 68 74 6d 6c 2c	61 70 70 6c 69 63 61 74	xt/html, applicat
0090	69 6f 6e 2f 78 68 74 6d	6c 2b 78 6d 6c 2c 61 70	ion/xhtml+xml,ap
00a0	70 6c 69 63 61 74 69 6f	6e 2f 78 6d 6c 3b 71 3d	plicatio n/xml;q=
00b0	30 2e 39 2c 69 6d 61 67	65 2f 77 65 62 70 2c 2a	0.9,imag e/webp,*
00c0	2f 2a 3b 71 3d 30 2e 38	0d 0a 55 70 67 72 61 64	/*;q=0.8 ..Upgrad
00d0	65 2d 49 6e 73 65 63 75	72 65 2d 52 65 71 75 65	e-Insecu re-Reque
00e0	73 74 73 3a 20 31 0d 0a	55 73 65 72 2d 41 67 65	sts: 1.. User-Age
00f0	6e 74 3a 20 4d 6f 7a 69	6c 6c 61 2f 35 2e 30 20	nt: Mozi lla/5.0
0100	28 57 69 6e 64 6f 77 73	20 4e 54 20 31 30 2e 30	(Windows NT 10.0
0110	2b 20 57 4f 57 26 24 20	20 41 70 70 6e 65 57 65	..WOW64) Applewe

Port numbers

IPv4 Header (Network Layer)

No.	Time	Source	Destination	Protocol	Length	Info
539	13.616736000	192.168.1.2	95.172.71.48	HTTP	794	GET / HTTP/1.1
561	14.028335000	192.168.1.2	95.172.71.48	HTTP	728	GET /wp-content/plugins/easy-fancybox/easy-fancybox.css.php?ver=1.3.4 HTTP/1.1
569	14.083458000	192.168.1.2	95.172.71.48	HTTP	727	GET /wp-content/uploads/jw-player-plugin-for-wordpress/player/jwplayer.js?ver=4.2.4 HTTP/1.1
623	14.340281000	95.172.71.48	192.168.1.2	HTTP	524	HTTP/1.1 200 OK (text/html)
634	14.346269000	192.168.1.2	74.125.130.95	HTTP	621	GET /css?family=Open+Sans:700,800,400 HTTP/1.1
644	14.405804000	95.172.71.48	192.168.1.2	HTTP	589	HTTP/1.1 301 Moved Permanently (text/html)

Frame 539: 794 bytes on wire (6352 bits), 794 bytes captured (6352 bits) on interface 0

Ethernet II, Src: HonHaiPr_6e:93:f7 (e0:06:e6:93:f7), Dst: Zte_49:31:3d (34:4b:50:49:31:3d)

Internet Protocol Version 4, Src: 192.168.1.2 (192.168.1.2), Dst: 95.172.71.48 (95.172.71.48)

Transmission Control Protocol, Src Port: 53062 (53062), Dst Port: 80 (80), Seq: 1, Ack: 1, Len: 740

Hypertext Transfer Protocol

GET / HTTP/1.1\r\nHost: www.inquirer.net\r\nConnection: keep-alive\r\nAccept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8\r\nUpgrade-Insecure-Requests: 1\r\nUser-Agent: Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/44.0.2403.157 Safari/537.36\r\nDNT: 1\r\nAccept-Encoding: gzip, deflate, sdch\r\nAccept-Language: en-US,en;q=0.8\r\n

0000	34 4b 50 49 31 3d e0 06	e6 9e 93 f7 08 00 45 00	4KPI1=...n...E.
0010	03 0c 60 73 40 00 80 06	2e f2 c0 a8 01 02 5f ac	..S@.....
0020	47 30 cf 46 00 50 b1 32	b9 81 32 27 c9 61 50 18	G0.F.P.2...'.aP.
0030	01 01 c8 ed 00 00 47 45	54 20 2f 20 48 54 54 50	GE T / HTTP
0040	2f 31 2e 31 0d 0a 48 6f	73 74 3a 20 77 77 77 2e	/1.1..Host: www.
0050	69 6e 71 75 69 72 65 72	2e 6e 65 74 0d 0a 43 6f	inquirer.net..Co
0060	6e 6e 65 63 74 69 6f 6e	3a 20 6b 65 65 70 2d 61	nnection : keep-a
0070	6c 69 76 65 0d 0a 41 63	63 65 70 74 3a 20 74 65	live..Ac cept: te
0080	78 74 2f 68 74 6d 6c 2c	61 70 70 6c 69 63 61 74	xt/html, applicat
0090	69 6f 6e 2f 78 68 74 6d	6c 2b 78 6d 6c 2c 61 70	ion/xhtml l+xml,ap
00a0	70 6c 69 63 61 74 69 6f	6e 2f 78 6d 6c 3b 71 3d	plicatio n/xml;q=
00b0	30 2e 39 2c 69 6d 61 67	65 2f 77 65 62 70 2c 2a	0.9,imag e/webp,*
00c0	2f 2a 3b 71 3d 30 2e 38	0d 0a 55 70 67 72 61 64	/*;q=0.8 ..Upgrad
00d0	65 2d 49 6e 73 65 63 75	72 65 2d 52 65 71 75 65	e-Insecu re-Reque
00e0	73 74 73 3a 20 31 0d 0a	55 73 65 72 2d 41 67 65	sts: 1.. User-Age
00f0	6e 74 3a 20 4d 6f 7a 69	6c 6c 61 2f 35 2e 30 20	nt: Mozi lla/5.0
0100	28 57 69 6e 64 6f 77 73	20 4e 54 20 31 30 2e 30	(windows NT 10.0
0110	28 57 69 6e 64 6f 77 73	20 4e 54 20 31 30 2e 30	(windows NT 10.0

Frame Header (Data Link Layer)

No.	Time	Source	Destination	Protocol	Length	Info
539	13.616736000	192.168.1.2	95.172.71.48	HTTP	794	GET / HTTP/1.1
561	14.028335000	192.168.1.2	95.172.71.48	HTTP	728	GET /wp-content/plugins/easy-fancybox/easy-fancybox.css.php?ver=1.3.4 HTTP/1.1
569	14.083458000	192.168.1.2	95.172.71.48	HTTP	727	GET /wp-content/uploads/jw-player-plugin-for-wordpress/player/jwplayer.js?ver=4.2.4 HTTP/1.1
623	14.340281000	95.172.71.48	192.168.1.2	HTTP	524	HTTP/1.1 200 OK (text/html)
634	14.346269000	192.168.1.2	74.125.130.95	HTTP	621	GET /css?family=Open+Sans:700,800,400 HTTP/1.1
644	14.405804000	95.172.71.48	192.168.1.2	HTTP	589	HTTP/1.1 301 Moved Permanently (text/html)



Frame 539: 794 bytes on wire (6352 bits), 794 bytes captured (6352 bits) on interface 0

Ethernet II, Src: NonHaiPr_6e:93:f7 (e0:06:e6:6e:93:f7), Dst: Zte_49:31:3d (34:4b:50:49:31:3d)

Internet Protocol Version 4, Src: 192.168.1.2 (192.168.1.2), Dst: 95.172.71.48 (95.172.71.48)

Transmission Control Protocol, Src Port: 53062 (53062), Dst Port: 80 (80), Seq: 1, Ack: 1, Len: 740

Hypertext Transfer Protocol

GET / HTTP/1.1\r\n

Host: www.inquirer.net\r\n

Connection: keep-alive\r\n

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8\r\n

Upgrade-Insecure-Requests: 1\r\n

User-Agent: Mozilla/5.0 (Windows NT 10.0; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/44.0.2403.157 Safari/537.36\r\n

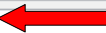
DNT: 1\r\n

Accept-Encoding: gzip, deflate, sdch\r\n

Accept-Language: en-US,en;q=0.8\r\n

Physical addresses

0000	34 4b 50 49 31 3d e0 06 e6 6e 93 f7 08 00 45 00	4KPI1=..n....E.
0010	03 0c 60 73 40 00 80 06 2e f2 c0 a8 01 02 5f ac	..s@...
0020	47 30 cf 46 00 50 b1 32 b9 81 32 27 c9 61 50 18	G0.F.P.2 ..2'.ap.
0030	01 01 c8 ed 00 00 47 45 54 20 2f 20 48 54 54 50	GE T / HTTP
0040	2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 77 77 77 2e	/1.1..Ho st: www.
0050	69 6e 71 75 69 72 65 72 2e 6e 65 74 0d 0a 43 6f	inquirer .net..Co
0060	6e 6e 65 63 74 69 6f 6e 3a 20 6b 65 65 70 2d 61	nnnection : keep-a
0070	6c 69 76 65 0d 0a 41 63 63 65 70 74 3a 20 74 65	live..Ac cept: te
0080	78 74 2f 68 74 6d 6c 2c 61 70 70 6c 69 63 61 74	xt/html, applicat
0090	69 6f 6e 2f 78 68 74 6d 6c 2b 78 6d 6c 2c 61 70	ion/xhtml+xml,ap
00a0	70 6c 69 63 61 74 69 6f 6e 2f 78 6d 6c 3b 71 3d	plicatio n/xml;q=
00b0	30 2e 39 2c 69 6d 61 67 65 2f 77 65 62 70 2c 2a	0.9,imag e/webp,*
00c0	2f 2a 3b 71 3d 30 2e 38 0d 0a 55 70 67 72 61 64	/*;q=0.8 ..Upgrad
00d0	65 2d 49 6e 73 65 63 75 72 65 2d 52 65 71 75 65	e-Insecu re-Reque
00e0	73 74 73 3a 20 31 0d 0a 55 73 65 72 2d 41 67 65	sts: 1.. User-Age
00f0	6e 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20	nt: Mozi lla/5.0
0100	28 57 69 6e 64 6f 77 73 20 4e 54 20 31 30 2e 30	(windows NT 10.0
0110	7b 30 57 4f 57 76 74 20 70 41 70 70 65 65 57 65	..WOW64) AppleWeb



References



- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.